

ECE 590.17: Lecture 5 – Markov Chain Monte Carlo

Graphical Models and Inference for Machine Learning

Duke University, Spring 2026

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Last Modified: 02/08/2026

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5.1 Why Markov Chain Monte Carlo?

Many inference problems require expectations with respect to the high-dimensional distribution

$$\mu(x) = \frac{1}{Z} e^{-\beta E(x)}, \quad Z = \sum_{x \in \mathcal{X}} e^{-\beta E(x)},$$

where $x \in \mathcal{X} = \mathcal{X}_0^n$ is a configuration (e.g., binary vectors, assignments, or codewords), $E(x)$ is a cost or negative log-likelihood, and β controls how sharply probability concentrates on low-cost states. In probabilistic terms, we can view $E(x)$ as a surrogate for $-\log p(x)$, so that $e^{-\beta E(x)}$ defines an unnormalized density. The desired tasks are:

- compute expectations $\mathbb{E}_\mu[f(X)]$,
- estimate marginal probabilities for components of X , and
- sample typical configurations from μ .

Direct summation is intractable because $|\mathcal{X}|$ grows exponentially with system size. Markov chain Monte Carlo (MCMC) replaces global summation by sampling: generate a Markov chain whose stationary distribution is μ , then approximate expectations by time averages.

Gibbs sampling

Gibbs sampling updates one variable at a time by drawing from its conditional distribution given all other variables. For a distribution $\mu(x)$ on $x = (x_1, \dots, x_n)$, the Gibbs update at index i samples x_i from $\mu(x_i | x_{-i}) \propto \mu(x)$. This defines a Markov chain with stationary distribution μ (assuming irreducibility). Glauber dynamics is Gibbs sampling specialized to binary variables on graphs. If there are configurations with probability zero, then Gibbs sampling can get trapped in regions surrounded by zero probability configurations.

Why Gibbs reaches μ (full support). If $\mu(x) > 0$ for all x , then each single-site conditional $\mu(x_i | x_{-i})$ is well-defined and strictly positive. The single-site Gibbs kernel at index i is

$$P_i(y | x) = \begin{cases} \mu(y_i | x_{-i}) & \text{if } y_{-i} = x_{-i}, \\ 0 & \text{otherwise.} \end{cases}$$

This satisfies detailed balance because, for x and y differing only at site i , we have

$$\begin{aligned} \mu(x)P_i(y | x) &= \mu(x_{-i})\mu(x_i | x_{-i})\mu(y_i | x_{-i})\mathbf{1}\{x_{-i} = y_{-i}\} \\ &= \mu(y_{-i})\mu(y_i | y_{-i})\mu(x_i | y_{-i}) \\ &= \mu(y)P_i(x | y). \end{aligned}$$

Choosing the update index uniformly gives a mixture kernel that is also reversible, hence μ is stationary. Positivity of all conditionals implies irreducibility, and self-loops occur with positive probability, so the chain is aperiodic and converges to μ .

Example: posterior inference. Suppose we observe data z and have a model $p(z|x)p(x)$. The posterior distribution $p(x|z)$ is proportional to $p(z|x)p(x)$, which fits the form above with $E(x) = -\log p(z|x) - \log p(x)$. MCMC provides a generic way to approximate posterior expectations when exact inference is infeasible.

5.2 Markov chains and detailed balance

Definition 1 (Finite-state Markov chain). A *finite-state Markov chain* (FSMC) with state space $\mathcal{X}$ (satisfying $|\mathcal{X}| < \infty$) is a sequence of random variables $X^{(0)}, X^{(1)}, X^{(2)}, \dots$ where each $X^{(t)} \in \mathcal{X}$ and the following Markov condition holds,

$$\mathbb{P}(X^{(t+1)} = y | X^{(t)} = x, X^{(0)}, X^{(1)}, \dots, X^{(t-1)}) = \mathbb{P}(X^{(t+1)} = y | X^{(t)} = x).$$

If $\mathbb{P}(X^{(t+1)} = y | X^{(t)} = x)$ does not depend on t , then the Markov chain is called *time invariant* (or homogeneous). When $\mathcal{X} = \mathcal{X}_0^n$ is a set of vectors, we write $X^{(t)} = (X_1^{(t)}, \dots, X_n^{(t)})$.

In this note, we assume the FSMC is time invariant and thus it is defined by the transition probabilities $P_{Y|X}(y | x)$ with

$$\mathbb{P}(X^{(t+1)} = y | X^{(t)} = x) = P_{Y|X}(y | x), \quad \sum_{y \in \mathcal{X}} P_{Y|X}(y | x) = 1.$$

For simplicity, we will also use the shorthand $P(y | x)$ and $Q(y | x)$ for Markov transition kernels. The t -step transition probabilities are

$$P^t(y | x) := \mathbb{P}(X^{(t)} = y | X^{(0)} = x),$$

so $P^1(y | x) = P(y | x)$. Using this, we see that the one-step transition probability associated with one-step of Gibbs sampling is given by

$$P(y | x) = \frac{1}{n} \sum_{i=1}^n P_i(y_i | x_{-i}).$$

A distribution π on $\mathcal{X}$ is stationary for $P(y | x)$ if

$$(\pi P)(y) := \sum_{x \in \mathcal{X}} \pi(x) P(y | x) = \pi(y).$$

A sufficient condition for stationarity (and reversibility) is *detailed balance*:

$$\pi(x) P(y | x) = \pi(y) P(x | y) \quad \text{for all } x, y.$$

This follows from summing both sides over x . If a chain is irreducible and aperiodic, it converges to its unique stationary distribution from any starting state. We will use π for the target distribution in general and set $\pi = \mu$ when the target is the Boltzmann distribution.

Irreducibility and aperiodicity. Irreducible means any state can be reached from any other state with positive probability in some number of steps. Aperiodic means the states cannot be partitioned into disjoint blocks where the block of future states is a deterministic function of the current block. Together, these ensure a single long-run distribution. In practice, most MCMC kernels are constructed to be irreducible by allowing small local moves that can eventually reach all configurations.

Balance vs detailed balance. Stationarity only requires the global balance condition $\pi P = \pi$, but detailed balance is a convenient sufficient condition because it is easy to check and often leads to simple acceptance rules.

5.3 Metropolis-Hastings

Given a proposal distribution $Q(y | x)$, the Metropolis-Hastings (MH) algorithm defines

$$P(y | x) = Q(y | x) \alpha(x, y), \quad \alpha(x, y) = \min \left\{ 1, \frac{\pi(y) Q(x | y)}{\pi(x) Q(y | x)} \right\}.$$

For target $\pi(x) \propto e^{-\beta E(x)}$, this becomes

$$\alpha(x, y) = \min \left\{ 1, \exp \left(-\beta(E(y) - E(x)) \right) \frac{Q(x | y)}{Q(y | x)} \right\}.$$

Detailed balance for MH. For $x \neq y$,

$$P(y | x) = Q(y | x) \alpha(x, y) = \min \left\{ Q(y | x), Q(x | y) \frac{\pi(y)}{\pi(x)} \right\}.$$

Multiplying by $\pi(x)$ gives

$$\pi(x) P(y | x) = \min \{ \pi(x) Q(y | x), \pi(y) Q(x | y) \},$$

which is symmetric in x and y , so

$$\pi(x) P(y | x) = \pi(y) P(x | y).$$

For $x = y$, detailed balance holds trivially. Thus, the Markov chain defined by MH is reversible and π is stationary.

Choosing a proposal. The proposal Q controls how the chain explores the state space. Local proposals (e.g., flip a single bit or spin) are simple and cheap but can mix slowly when configurations are highly correlated. Larger proposals can move faster but may have low acceptance. Good performance requires balancing acceptance rate and exploration.

Independence sampler. If $Q(y | x) = q(y)$ is independent of x , then MH becomes an independence sampler. It can be efficient when q is close to π , but can fail badly if q misses important regions of the target.

Metropolis rule. If Q is symmetric, $Q(y | x) = Q(x | y)$, then

$$\alpha(x, y) = \min\{1, \exp(-\beta(E(y) - E(x)))\}.$$

Moves that decrease energy are always accepted; increases are accepted with probability $e^{-\beta\Delta E}$.

5.4 Glauber dynamics

Glauber dynamics is the name of the single-site Gibbs sampling update rule often used for Ising models. For a binary spin system $\sigma \in \mathcal{X} = \{\pm 1\}^N$ with distribution $\mu_\beta(\sigma) \propto e^{-\beta E(\sigma)}$, pick a site i uniformly at random and update σ_i conditioned on the neighbors:

$$\mathbb{P}(\sigma_i = +1 | \sigma_{-i}) = \frac{1}{1 + \exp(-\beta \Delta_i E)}, \quad \Delta_i E = E(\sigma^{i \rightarrow -1}) - E(\sigma^{i \rightarrow 1}), \quad (5.1)$$

where $\Delta_i E$ is the energy change caused by flipping $\sigma_i = 1$ to $\sigma_i = -1$ with all other spins fixed. This update satisfies detailed balance and yields the same stationary measure as the Metropolis rule. This is Gibbs sampling for a single site, which is why Glauber dynamics appears as the MCMC workhorse for binary graphical models.

Conditional distribution view. Glauber dynamics can be seen as repeatedly sampling one variable from its conditional distribution given all others. This is a special case of Gibbs sampling, and it naturally targets the desired joint distribution.

Comparison with Metropolis. For binary variables, both Metropolis and Glauber require only local energy differences. Glauber updates always accept a new value drawn from the correct conditional, while Metropolis proposes a specific move and may reject it. The two methods often behave similarly but can differ in autocorrelation and convergence rates.

5.5 MCMC estimators and convergence

Given samples $x^{(1)}, \dots, x^{(T)}$ from a Markov chain with stationary distribution π , we estimate expectations by

$$\mathbb{E}_\pi[f(X)] \approx \frac{1}{T} \sum_{t=1}^T f(x^{(t)}).$$

The error is controlled by the chain's *mixing time* and *autocorrelation*. In practice:

- Run a *burn-in* period to reach stationarity.
- Thin or subsample if autocorrelations are high.
- Monitor convergence via trace plots, autocorrelation, or multiple chains.

Free energy barriers. If the energy landscape has multiple basins separated by a high barrier, MCMC can take exponential time to move between them. This can lead to metastability and long autocorrelation times.

Autocorrelation and effective sample size. Let $Y_t = f(X^{(t)})$. The empirical average has variance

$$\text{Var}\left(\frac{1}{T} \sum_{t=1}^T Y_t\right) \approx \frac{\sigma_f^2}{T} \left(1 + 2 \sum_{k=1}^{\infty} \rho_k\right),$$

where (assuming stationarity) $\sigma_f^2 = \text{Var}(Y_t)$ and the lag- k autocorrelation is

$$\rho_k = \frac{\text{Cov}(Y_t, Y_{t+k})}{\text{Var}(Y_t)}.$$

In the first equation, the RHS term in parentheses defines the *integrated autocorrelation time*. The effective sample size is T divided by this factor, so high autocorrelation can make a long run behave like only a few independent samples.

Burn-in and initialization. Starting from a highly structured state (e.g., all +1 spins) can bias early samples. A burn-in period discards the initial segment of the chain to reduce this effect. In practice, one checks sensitivity by running multiple chains from diverse initial states.

Mixing time and convergence. For an irreducible, aperiodic finite chain, $P^t(x, \cdot) \rightarrow \pi$ as $t \rightarrow \infty$ for any start state x . Define the total variation distance

$$d(t) = \max_{x \in \mathcal{X}} \|P^t(\cdot | x) - \pi(\cdot)\|_{\text{TV}},$$

where $\|\nu(\cdot) - \mu(\cdot)\|_{\text{TV}} = \frac{1}{2} \sum_{x \in \mathcal{X}} |\nu(x) - \mu(x)|$ and the mixing time is

$$t_{\text{mix}}(\varepsilon) = \min\{t : d(t) \leq \varepsilon\}.$$

Smaller t_{mix} means faster convergence; bottlenecks (e.g., free-energy barriers) increase it by slowing communication between regions of high probability.

5.6 The 2D Ising model

We now specialize to the 2D Ising model on a square lattice, but we describe it in probabilistic language. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be an $L \times L$ grid with vertices at intersections and edges between adjacent vertices. We assume periodic boundary conditions and each vertex $i \in \mathcal{V}$ carries a binary variable $\sigma_i \in \{\pm 1\}$. The energy is

$$E(\sigma) = - \sum_{(i,j) \in \mathcal{E}} J \sigma_i \sigma_j - h \sum_{i \in \mathcal{V}} \sigma_i,$$

with coupling $J > 0$ for the ferromagnetic case and external field h . The Boltzmann distribution is

$$\mu_\beta(\sigma) = \frac{1}{Z(\beta)} \exp\left(\beta J \sum_{(i,j) \in \mathcal{E}} \sigma_i \sigma_j + \beta h \sum_{i \in \mathcal{V}} \sigma_i\right).$$

Probabilistic interpretation. When $J > 0$, configurations with neighboring spins equal are more likely. This model is equivalent to a pairwise Markov random field with compatibility functions $\psi_i(\sigma_i) = \exp(\beta h \sigma_i)$, $\psi_{ij}(\sigma_i, \sigma_j) = \exp(\beta J \sigma_i \sigma_j)$. Thus, the Ising model is a simple, structured distribution over binary images.

Local conditional distribution. Given neighbors of i , the effective field is given by

$$h_i^{\text{eff}} = h + J \sum_{j \in \partial i} \sigma_j. \quad (5.2)$$

Then, the conditional probability of σ_i is logistic in h_i :

$$\mathbb{P}(\sigma_i = +1 \mid \sigma_{-i}) = \frac{1}{1 + \exp(-2\beta h_i^{\text{eff}})}. \quad (5.3)$$

Since $\Delta_i E(\sigma) = 2h_i^{\text{eff}}$, we recover Glauber dynamics (5.1) which is a good test case for MCMC.

Observables. Two standard observables are the magnetization and energy per spin:

$$M(\sigma) = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} \sigma_i, \quad U(\sigma) = \frac{E(\sigma)}{|\mathcal{V}|}.$$

The model exhibits a sharp change in behavior at a known critical temperature $T_c = 1/\beta_c$; below T_c configurations concentrate on aligned patterns, while above T_c spins decorrelate quickly. We focus on these observable changes rather than the exact formula. For $J = 1$ on a large grid, it can be shown that $T_c \approx 2.27$ (i.e., $\beta_c \approx 0.441$).

What do these mean without physics? Magnetization is simply the average of the binary variables. At high temperature (small β), the distribution is close to uniform and magnetization concentrates near zero. At low temperature (large β), the distribution favors configurations where most variables agree, and magnetization concentrates near ± 1 .

5.7 Implementing MCMC for the 2D Ising model

We outline a practical implementation using single-spin Glauber or Metropolis updates.

5.7.1 State and energy changes

Represent the lattice as an $L \times L$ array of spins. For a given site i , compute the effective field h_i^{eff} using (5.2) from the four neighbors. Then, update σ_i using (5.3). This approach shows each update has $O(1)$ complexity.

Boundary conditions. Periodic boundary conditions identify the left and right edges, and the top and bottom edges, so every site has exactly four neighbors. This reduces edge effects and better reflects the large-lattice behavior. If open boundaries are used instead, the update rule is the same but the neighbor set has fewer elements at the edges.

Initialization. Two common initializations are a random state (each spin ± 1 with equal probability) and an ordered state (all $+1$ or all -1). Random starts are useful for exploring high-temperature behavior; ordered starts can expose long mixing times at low temperature.

5.7.2 Glauber update

For each step:

1. Pick a site i uniformly at random.
2. Compute h_i^{eff} .
3. Set $\sigma_i = +1$ with probability $1/(1 + e^{-2\beta h_i^{\text{eff}}})$ and -1 otherwise.

Glauber updates often mix more smoothly near criticality but are comparable in cost to Metropolis.

Pseudocode

```
for sweep = 1..T:
  for k = 1..L^2:
    pick i uniformly
    compute heff_i = h + J * sum_{j in N(i)} sigma_j
    if rand() < 1/(1+exp(-2\beta * heff_i)):
      sigma_i = 1
    else:
      sigma_i = -1
  measure observables
```

Implementation detail. For reproducibility, fix a random seed and report it with results. A typical burn-in for $L = 32$ is 500–1000 sweeps at high temperature and longer near criticality; use trace plots to choose an adequate value.

5.7.3 Metropolis update

For each step:

1. Pick a site i uniformly at random.
2. Compute $\Delta_i E = 2\sigma_i h_i^{\text{eff}}$.
3. If $\Delta_i E \leq 0$, flip σ_i .
4. Otherwise, flip with probability $e^{-\beta \Delta_i E}$.

One *sweep* is $|\mathcal{V}|$ such updates.

5.7.4 Measuring statistics

At the end of each sweep, compute $M(\sigma)$ and $U(\sigma)$ and store them. To estimate expectations, discard a burn-in portion and average over the remaining samples. Plotting M and U versus sweep index is a simple diagnostic for convergence.

5.7.5 An illustrative experiment

Choose $L = 32$ and $J = 1$. Then, simulate for $T = 5000$ sweeps. Run at three temperatures: $T = 1.5$ (low), $T = 2.3$ (near critical), and $T = 3.5$ (high). Compare the time series of M and the autocorrelation function of M . You should see slow mixing near $T \approx T_c$, a stable magnetized state at low T , and fast fluctuations around zero at high T .

5.8 What results to expect

Low temperature ($T < T_c$). The system develops spontaneous magnetization. Starting from a random state, the chain rapidly coarsens into large domains and eventually settles into a state with $|M|$ close to 1 (up to finite-size effects). MCMC can get trapped in one of the two magnetized phases for a long period because flipping the global magnetization requires crossing a large free-energy barrier. Equivalently, the distribution has two dominant modes separated by low-probability configurations, so local updates rarely switch between them.

From a probabilistic standpoint, there are two dominant modes centered near all-+1 and all-−1 configurations. Local MCMC updates move within one mode quickly, but jumping between modes is rare, so long runs may appear biased toward one sign of M .

High temperature ($T > T_c$). Spins decorrelate quickly. The magnetization concentrates near zero and energy fluctuations are modest. Mixing is fast because local changes are rarely penalized by large energy barriers.

In probabilistic terms, μ becomes close to a product distribution with weak pairwise correlations. Local updates explore the space efficiently because there are no deep modes that trap the chain.

Near criticality ($T \approx T_c$). Critical slowing down occurs: correlation lengths become large and MCMC mixing time increases. Autocorrelation times can grow polynomially with system size, and careful sampling is required.

You should expect large clusters of aligned spins to form and dissolve over long timescales. This produces slow decay in the autocorrelation of observables and makes convergence diagnostics essential.

Finite-size effects. For finite L , the phase transition is smoothed. One typically studies $|M|$ and its susceptibility across T , and observes sharper transitions as L increases.

5.9 Diagnostics and practical tips

- Use multiple chains with different initializations to assess convergence.
- Track running averages of M and U ; stable plateaus indicate stationarity.
- Estimate autocorrelation times to set effective sample sizes.
- Near T_c , consider cluster methods (e.g., Swendsen-Wang or Wolff) to reduce critical slowing down.

Interpreting trace plots. A trace plot of M should fluctuate around a stable mean once the chain has mixed. If the trace drifts slowly or remains stuck at high absolute values, the chain may be in a single mode and not exploring the full distribution.

Acceptance rates. For Metropolis updates, extremely high acceptance can indicate too small steps (slow exploration), while extremely low acceptance can indicate overly aggressive proposals. Single-spin flips typically yield moderate acceptance in the Ising model; this is one reason they are widely used.

5.10 Connections to optimization and inference

MCMC provides approximate marginals for probabilistic inference and approximate solutions for optimization problems via annealing. For combinatorial optimization, one often defines $E(x)$ as a cost function and runs a simulated annealing schedule, slowly increasing β to bias the chain toward low-energy configurations. This idea appears in K-SAT, MAX-SAT, and decoding, linking MCMC to the broader themes of the course.

Simulated annealing intuition. At small β , the chain explores widely; at large β , it concentrates on low-cost configurations. A gradual increase of β can help avoid early trapping while still guiding the chain toward good solutions. In practice, the schedule must be tuned to balance runtime and solution quality.

Relation to BP. Belief propagation provides approximate marginals by deterministic messages, while MCMC provides stochastic estimates. In some models, BP is faster but less robust when the graph has short loops or strong couplings. MCMC can be slower but is more broadly applicable. Comparing the two gives insight into algorithmic tradeoffs.